
NH33B-09: Determination of a flooding rate using GIS and water-rise simulations in a LiDAR bare-earth model Paul Weinstein, & Antonios E. Marsellos

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Progressive flooding depends on the rate at which an area will be completely flooded. A flat area such as of this in Long Beach in New York is prone to quick flooding due to morphology. Public officials and authorities require a map showing regions of flooding rate, that is what is the estimated rate that an area will be completely flooded, so they may focus on those calls that require an immediate response. The analysis of flood risk has been widely studied in this area, however, there has not been much study on how fast a flood will develop considering features on the surface. There is a need for an estimated time window at which a flood process will take place, which is termed as the flood window. The flood window is the function of an area being flooded in response to the elevation of water level upon a flood. The first derivative of the flood window function will provide the rate at which the area will be flooded. The flood window is dependent on the slope of the area. A steep slope measurement causes the area to have a wide flood window. A gentle slope will result in a fast rate of flooding and creates a narrow flood window that allows less time for responders to react. A maximum value of the flood rate was determined for each cell on a 100 m square grid. The simulations were developed using a LiDAR-DEM in a GIS software such as Global Mapper to consider surface features as obstacles against water flow accumulation, and to simulate flooding of Long Beach at various water heights. Approximately 80-water-rise simulations were performed in Global Mapper in intervals of .1 meters for a range of sea level up to 8 meters at which the bare earth model of the study area is completely flooded. Regardless to the elevation the provided flood rate map shows the estimated available time of response of the officials against an imminent flood hazard. This method may be applied in other regions as well, and it provides significant information to assist first responders in concentrating on those areas that will be first at risk.

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